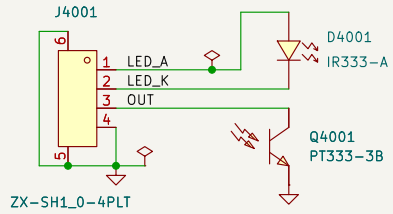
OOMWOO cliff sensor board (rev 1, 2026-10-28). One design, fitted 6x. No lenses: narrow-beam 5 mm parts (IR333-A 20 deg, PT333-3B black lens) tilted toward each other so their beams cross at the floor. The tilt sets the reference distance: crossing depth = (spacing / 2) / tan(tilt), e.g. 18 mm spacing: 30 deg -> 15.6 mm, 25 deg -> 19.3 mm, 20 deg -> 24.7 mm. Beyond the crossing point the beams diverge, so a far floor (a cliff) returns little light even if it is white.
The main board switches the LED cathode (CLIFF_LED_EN, modulate it to cancel ambient light) and pulls OUT up (47k).
Refs in the 4000s: each board has its own range.



Assembly (JLC): D4001 and Q4001 THT with the ASSEMBLY STANDOFF (~2 mm) so the leads can be bent; pin bending to the tilt angle, or bend by hand in the 3D-printed housing. LED: flat side = cathode. Q8501: flat side / square pad = emitter. Spacing D4001-Q4001: 18 mm centre to centre.

- VLED PWM period <<15 usec
- duty cycle ~10%

OOMWOO cliff sensor

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Title: Cliff sensor board (fit 6x)

Size: USLetter	Date: 2026-10-28	**Rev: 1**
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